

Freescalé's Symphony™ Class D Digital Amplifier Chipset for Car Audio and Navigation Systems (AA108)

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Freescal Audio Solutions Overview

Provide High-quality Programmable Audio Solutions
to the Consumer & Automotive Customers.

Comprehensive Solution

Hardware

Aggressive Roadmap

- Symphony™ family of 24bit DSP's for digital audio applications.
- **Cutting edge technology for Digital Amplification with advanced feedback system.**

Software

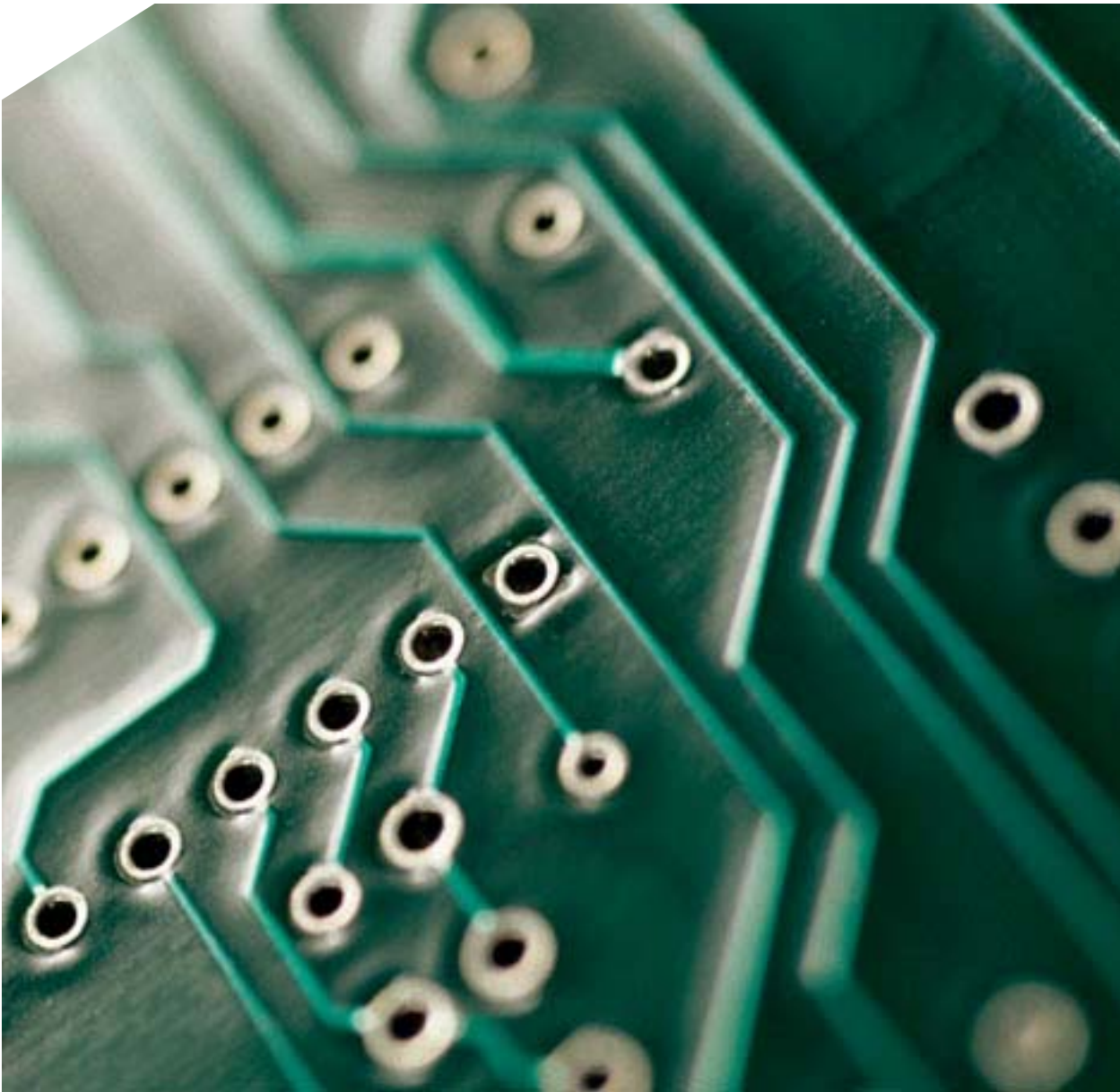
Develop & Share Extensive Audio Algorithm Library

- Always working on the latest Software IP in the Audio industry.
- Plug and Play architecture allow for quick and easy software and post processing algorithm integration.
- 3rd party developers are using the Freescale architecture.

System

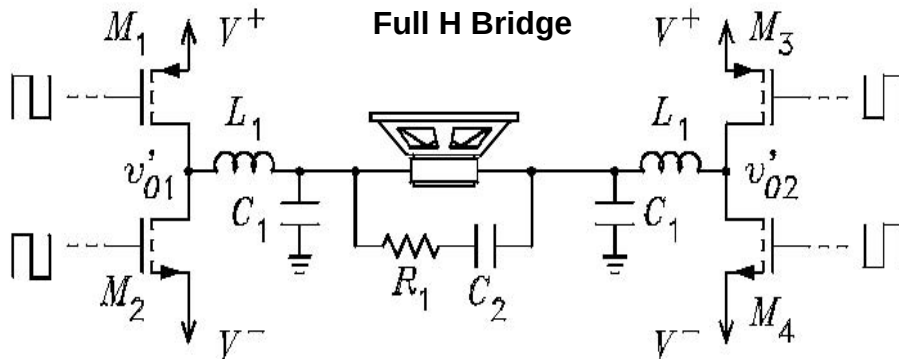
Reference Designs

- Symphony™ Surround Module
- DH1 (Dolby Headphone)
- **Symphony™ Class D Amplifier**

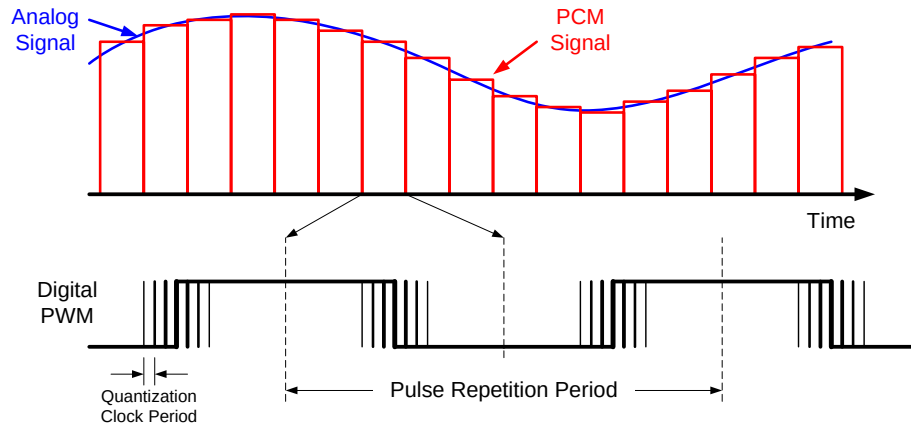


Freescal Digital Amplifier System Overview

Class D Switching Amplifier



Class D



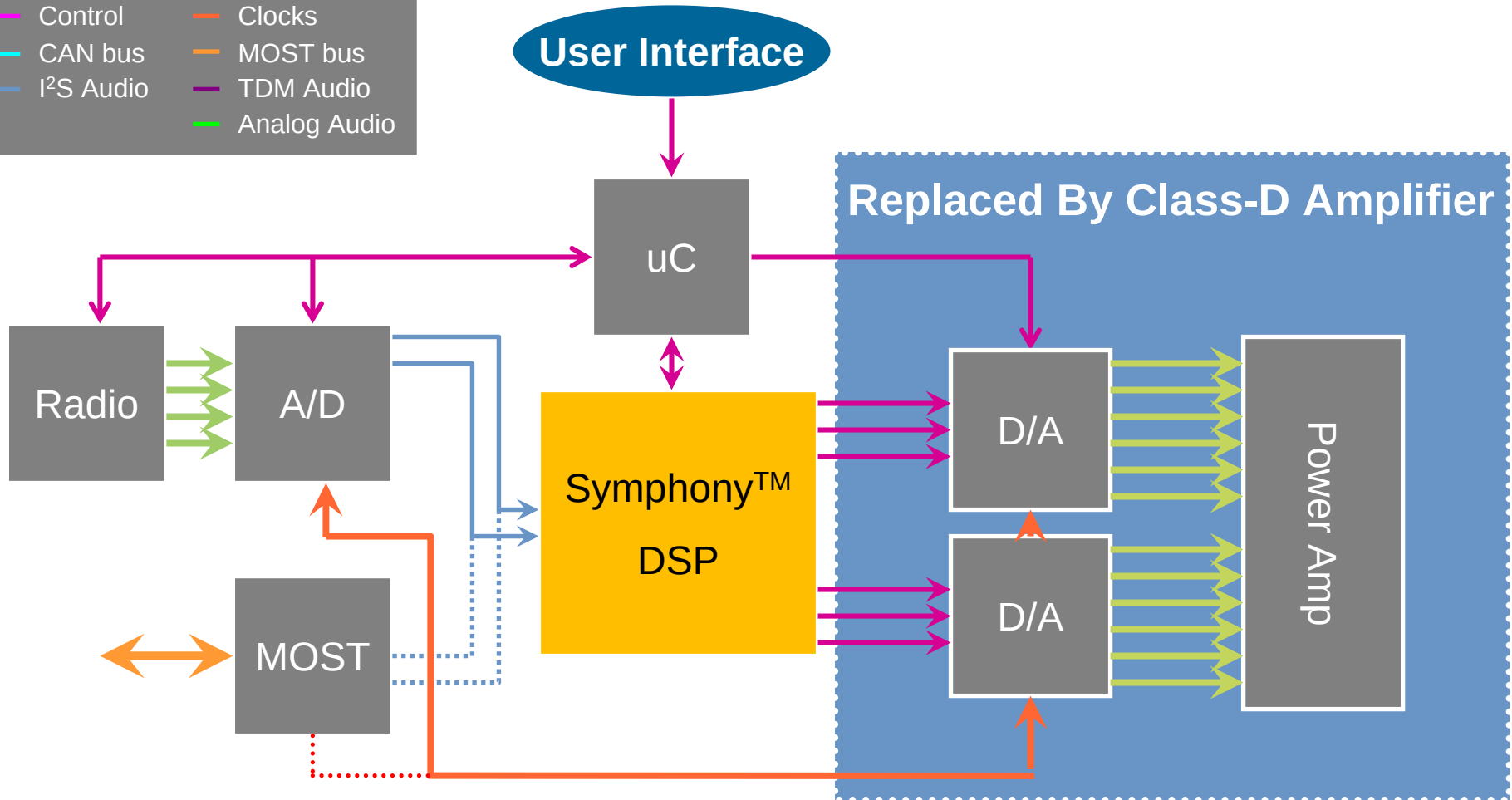
Topology	Half-bridge (simpler) and full-bridge (better in audio performance) configurations
Power Efficiency	90%~95%
EMI	Similar to other switching applications
Audio Quality	Better
Form Factor*	Smallest
System Cost*	Savings in heatsink, additional output filter cost, EMI shielding cost
Applications - Today	Subwoofers, DVD Receivers, High-wattage systems, Aftermarket automotive, limited OEM automotive
Applications - Future	Everywhere
* Result of Power Efficiency	

* Result of Power Efficiency

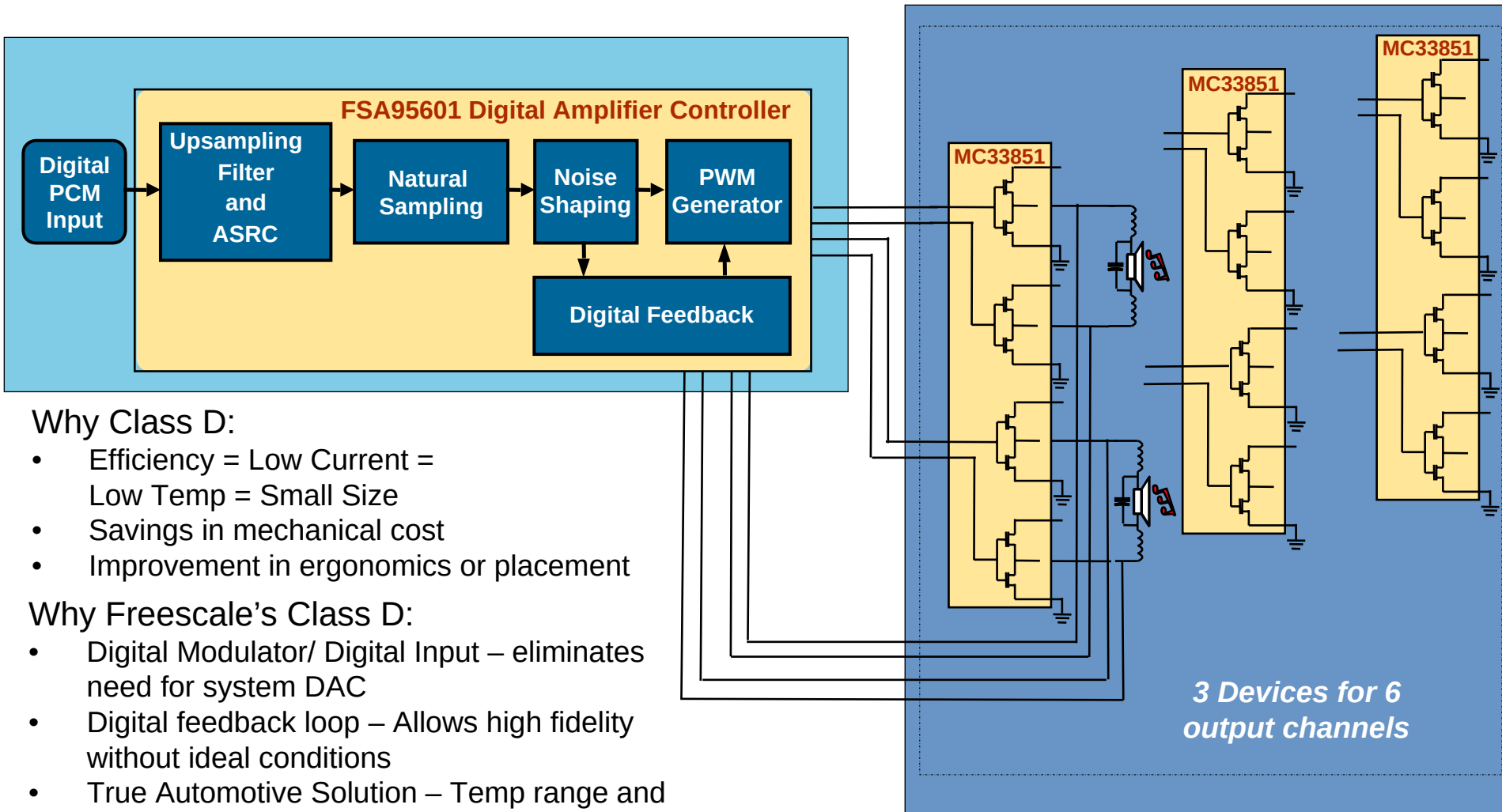
Car Audio Amplifier System

Key

Control	Clocks
CAN bus	MOST bus
I ² S Audio	TDM Audio
Analog Audio	



Freescal Digital Amplifier System



Why Class D:

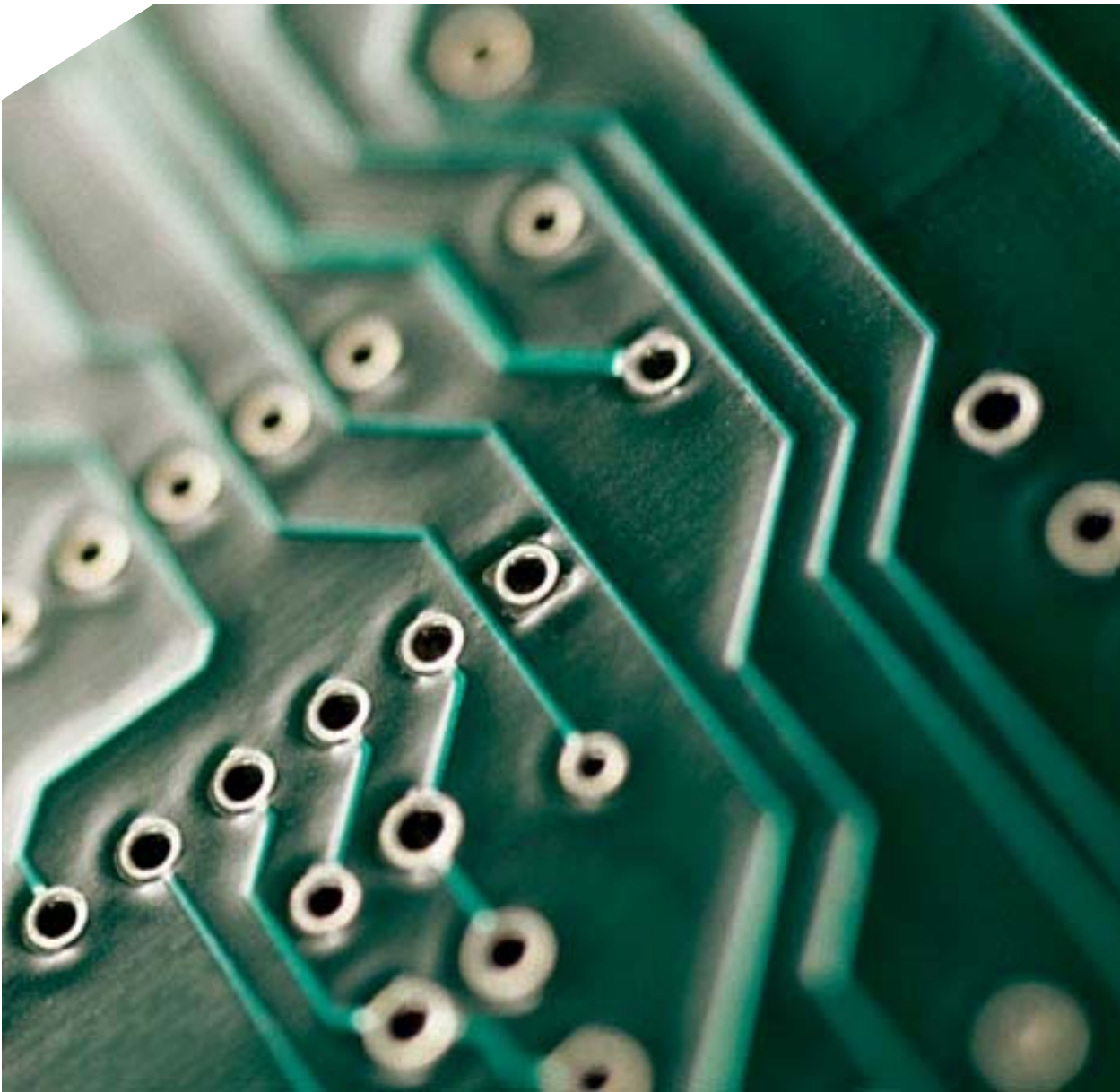
- Efficiency = Low Current = Low Temp = Small Size
- Savings in mechanical cost
- Improvement in ergonomics or placement

Why Freescale's Class D:

- Digital Modulator/ Digital Input – eliminates need for system DAC
- Digital feedback loop – Allows high fidelity without ideal conditions
- True Automotive Solution – Temp range and diagnostic features provided

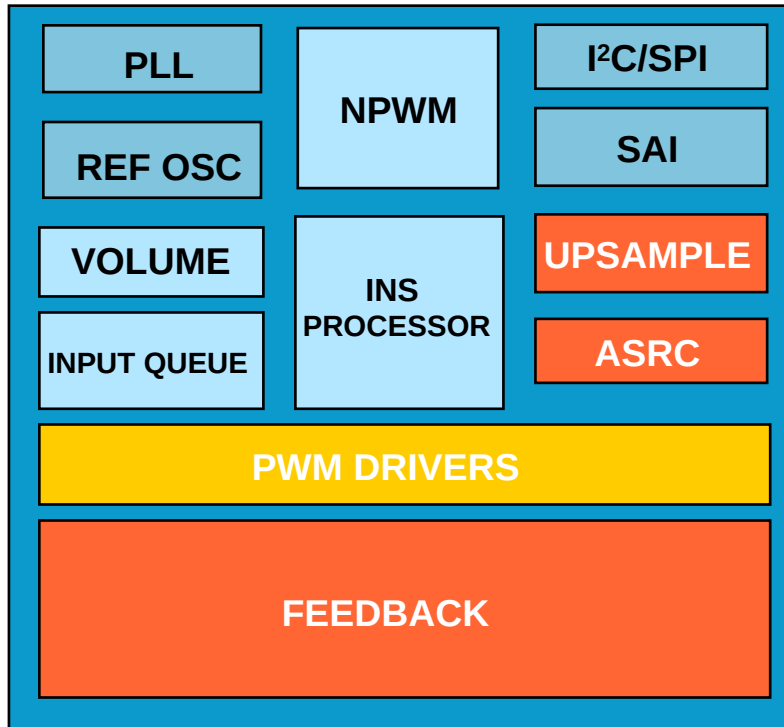
Freescalé's Unique Value In Digital Amplifier Solutions

- **Freescalé's Digital Amplifier solution is optimized for cost/performance:**
 - Increased power output with lower heat dissipation allowing more channels in smaller form factors
 - Superior power supply noise rejection via high performance digital feedback that substantially reduces power supply costs
 - Outstanding SNR and THD
 - Substantially reduced EMI
 - Advanced diagnostics & protection
 - Sound fidelity comparable or better to existing analog designs
- **Freescalé offers full digital audio portfolio from the signal processing front-end to the power stage back-end:**
 - Industry leading Symphony™ digital signal processors – DSP563xxx
 - Digital Amplifier Modulator (unique design high performance digital feedback) – FSA95601
 - Integrated MOSFET Power Stage (advanced diagnostics & protection) - MC33851
- **Freescalé has established market presence in Automotive, Consumer and Professional Multi-Channel Digital Audio Processing**



Freescal Digital Amplifier System Overview

FSA95601 Digital Amplifier Controller

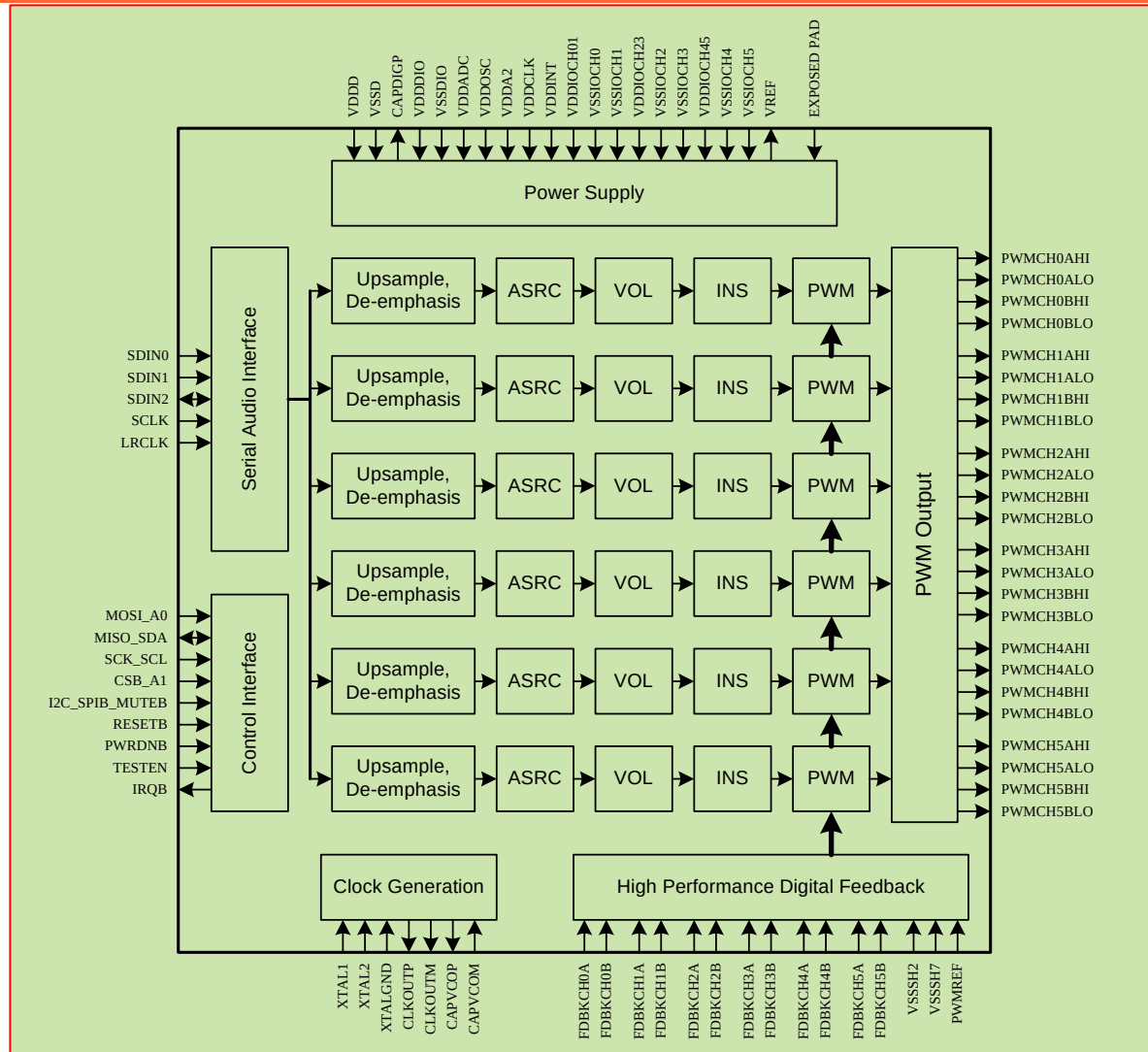


Status:

- Working prototype samples available now
- Production Samples in 4Q06
- EVM board operational
- FSA95601 production chip qualified 1Q07

- Digital Amplifier Controller
 - **Six audio channels**
 - Converts PCM inputs to PWM outputs
 - Drives external half-bridge or H-bridge power stages
- High Performance Audio Amplifier
 - Low distortion (typical THD < 0.01%, 20-20KHz, with 1KHz, -3dB FS (12W) input, 14.4V power supply, 4 Ohm load)
 - Dynamic range > 100dB, unweighted
 - Patented high performance digital feedback technology corrects for power stage non-idealities and rejects power supply noise while maintaining superior audio performance
- Advanced PWM Features
 - Programmable break-before-make
 - Independent modulation of each two-sided PWM edge
 - Digital dither
 - Conventional PWM and suppressed carrier PWM operating modes
- Simple System Interface Features
 - On-chip asynchronous sample rate converter (ASRC) accommodates PCM inputs from 32kHz to 192kHz
 - I²C or SPI slave port for control interface
 - Serial Audio Interface (SAI) supporting I²S, TDM, and other common audio formats up to 192 KHz input
 - On-chip crystal oscillator and low-jitter PLL
- 80-pin Exposed Pad TQFP Package

FSA95601 Block Diagram



MC33851 Dual H-Bridge Power Stage

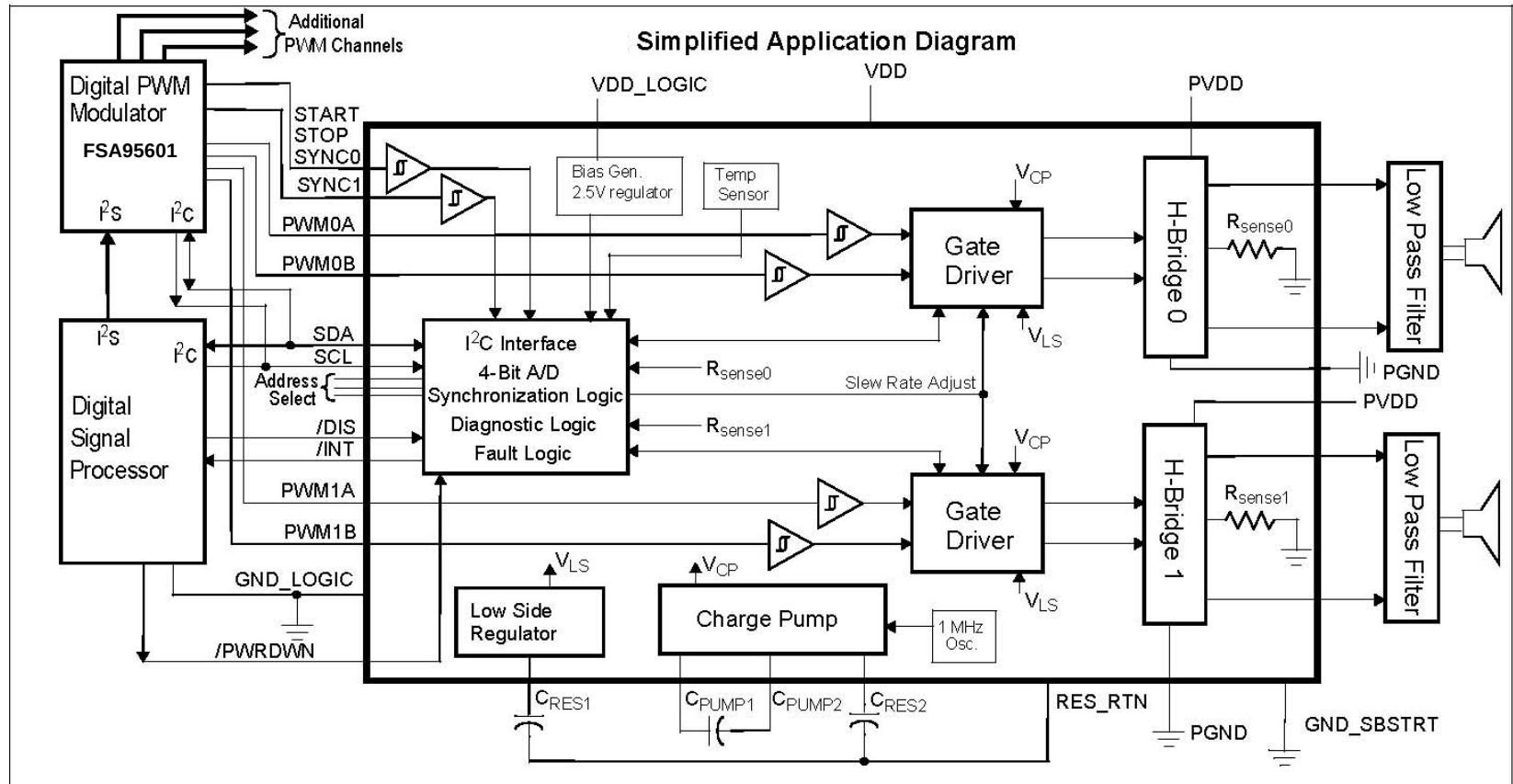
Features

- Optimized for automotive sound systems and harsh environments
- Dual 50W Power H-bridges (Outputs can be bridged to 100W)
- Shoot-thru (cross-conduction) prevention circuitry
- Thermal over-temperature, over-current and under-voltage pre-warning reporting via I²C
- Speaker power level reporting via I²C
- Variable slew rate control to manage EMI via I²C
- Short-Circuit Protected Outputs
- Under-voltage, over-voltage and over-temperature shutdown
- Eight I²C addresses available
- Low R_{DS(on)} (0.1 Ohm, 1 A at 25°C)
- 5.0 and 3.3 V Logic compatibility
- Audio pop-suppression circuitry

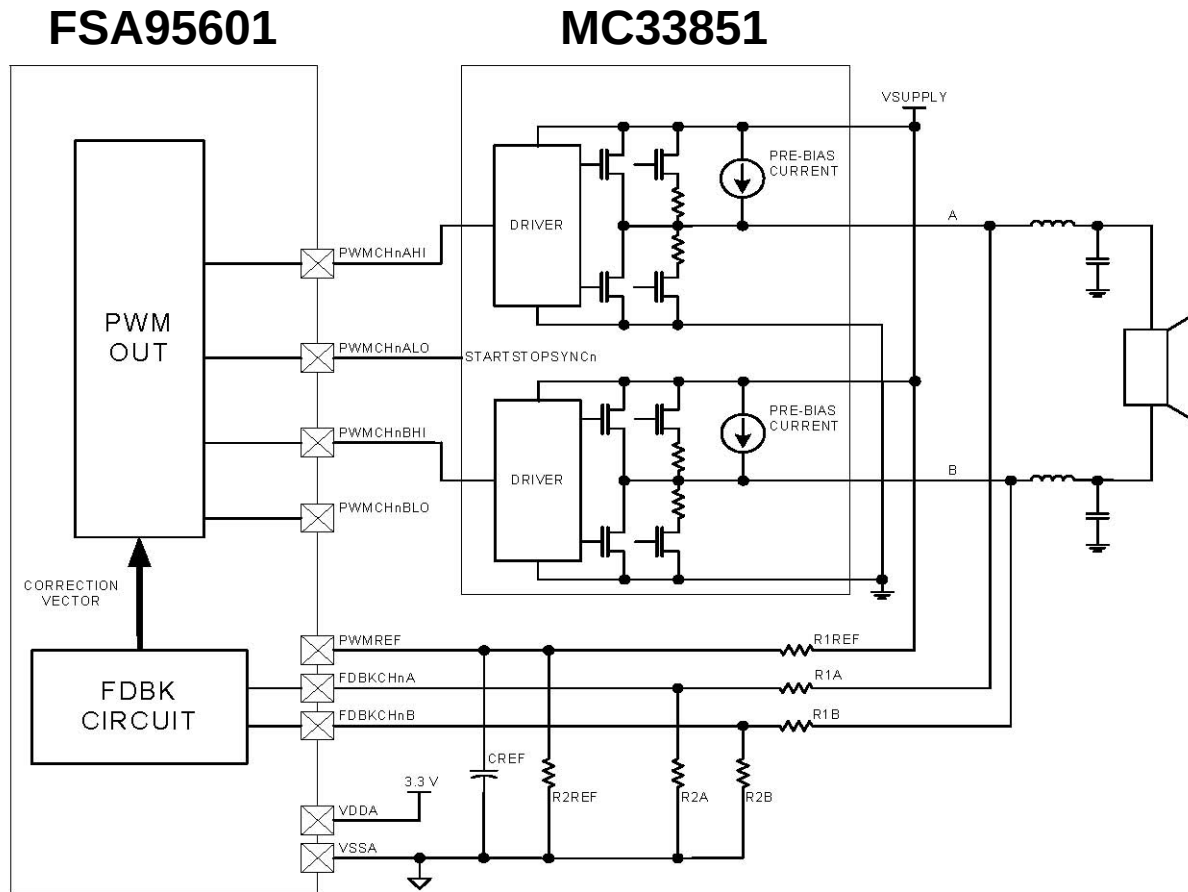
MC33851 Dual H-Bridge Power Stage

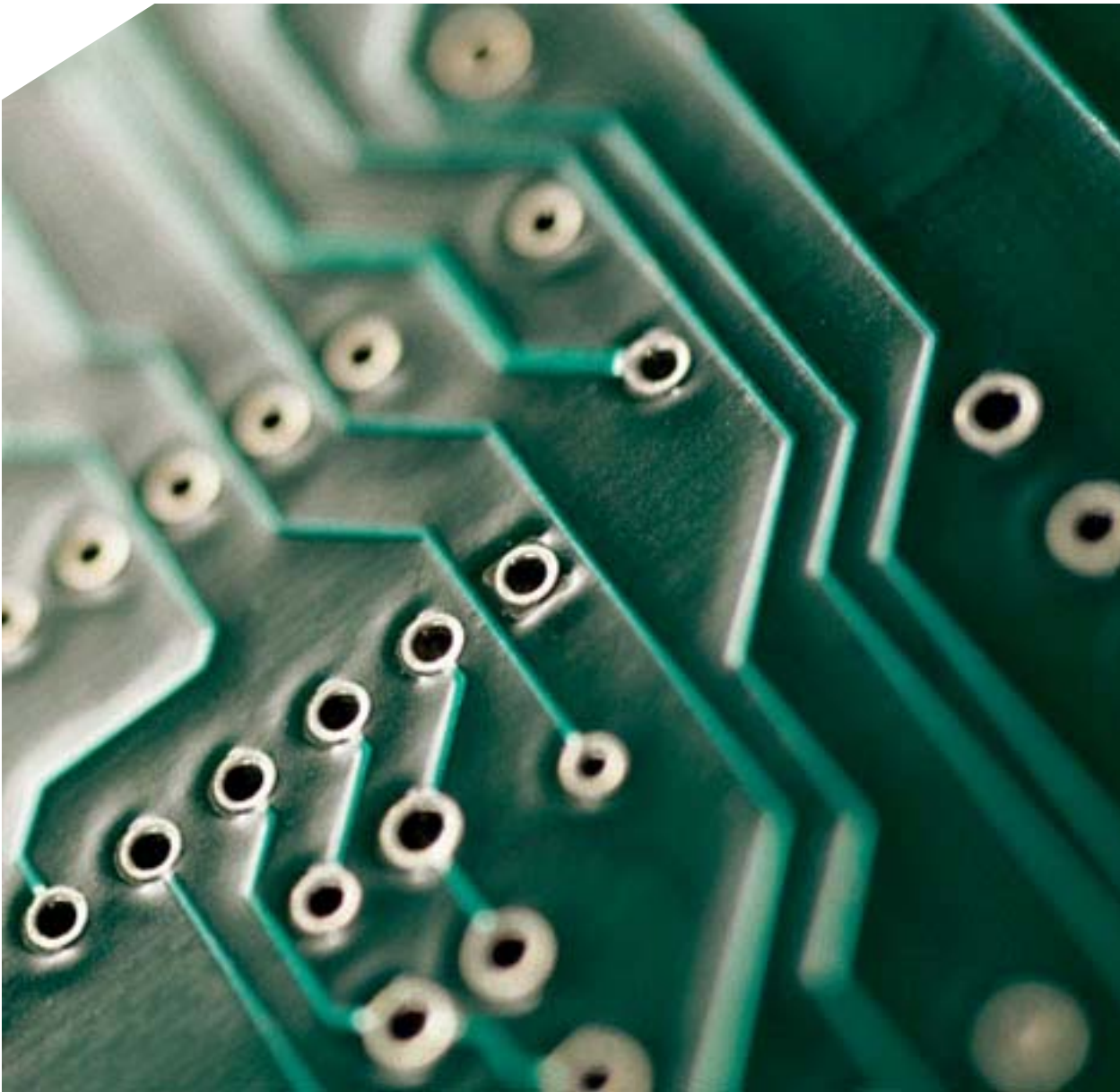
- MC33851 is designed to be used as the output power stage for applications requiring 50W to 100W per channel
- MC33851 is designed to interface with the FSA95601 Symphony Digital Amplifier Controller
- Power MOSFET Switching Speed Control reduces EMI with no audio quality degradation, when used with the FSA95601
- Advanced turn-on/off audio click and pop suppression when used in conjunction with the FSA95601
- Advanced protection and diagnostics features.
- Automotive qualification

MC33851 Application Diagram



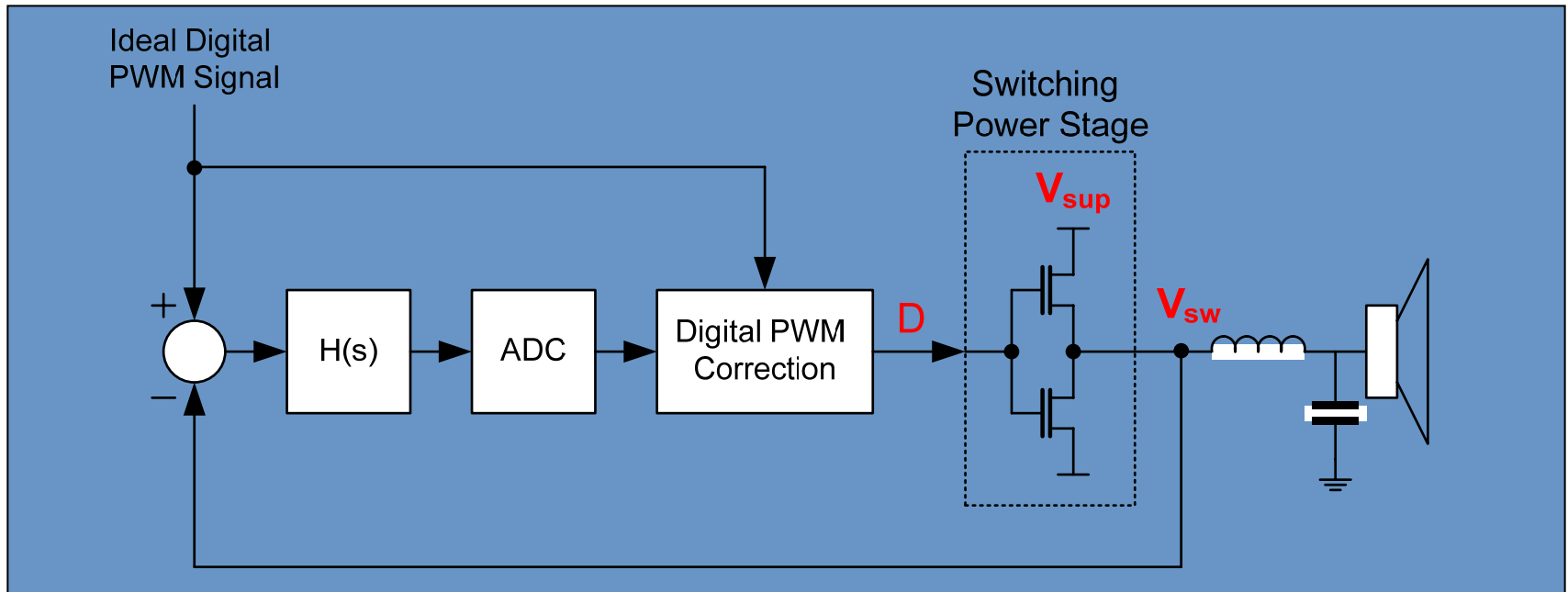
FSA95601 and MC33851 Interface





Advantages of the Freescale High Performance Digital Feedback

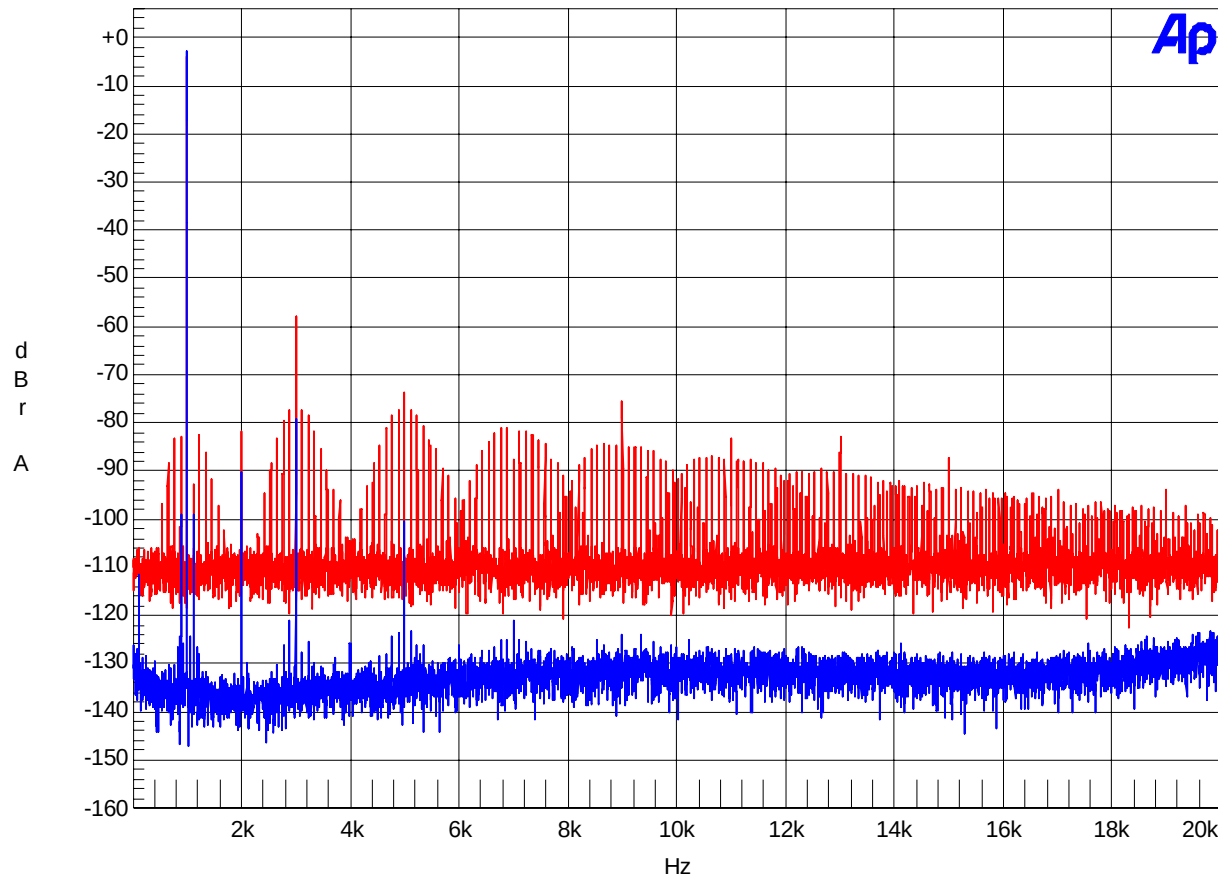
FSA95601 High Performance Digital Feedback



One of the key advantages of the Freescale digital amplifier solution

- **Feedback not common in competitor's solution but even when present not as good a result as with the patented Freescale method**
- **Corrects power stage non-idealities**
- **Provides superior power supply noise rejection. Enables use of lower cost power stage and power supply components**
- **Allows slower switching of the power FETs to reduce EMI without audio performance penalty**

FSA95601 Feedback Benefits: Excellent Power Supply Noise Rejection



Input signal:
1 kHz @ -3dBFS

Supply: 14.4 V

Power Supply
Noise:
1 Volt peak-peak
120 Hz

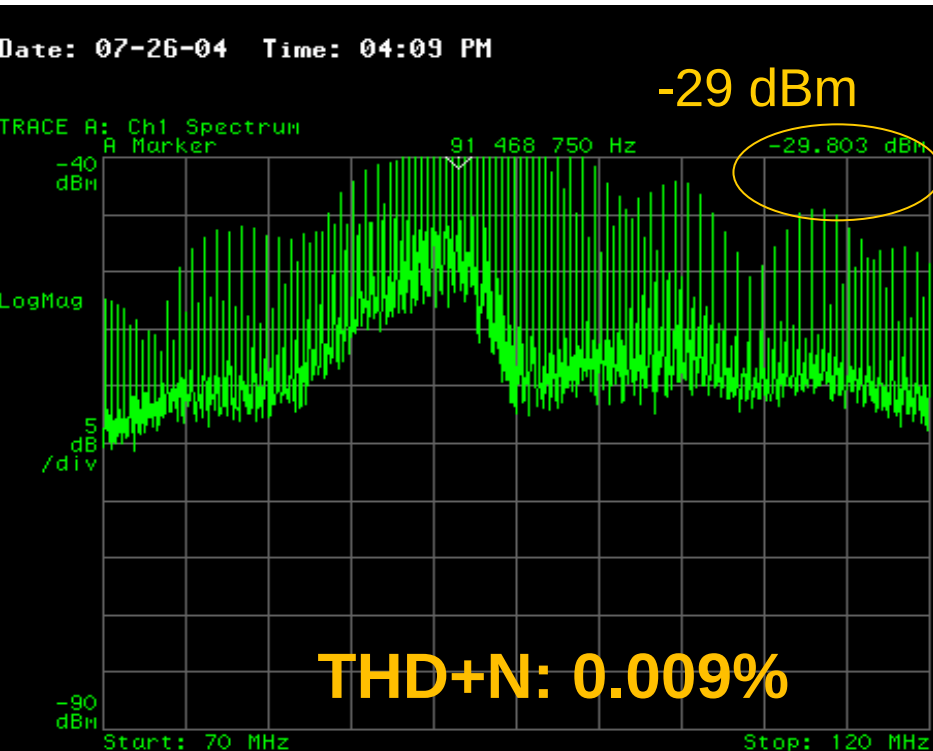
no feedback

with feedback

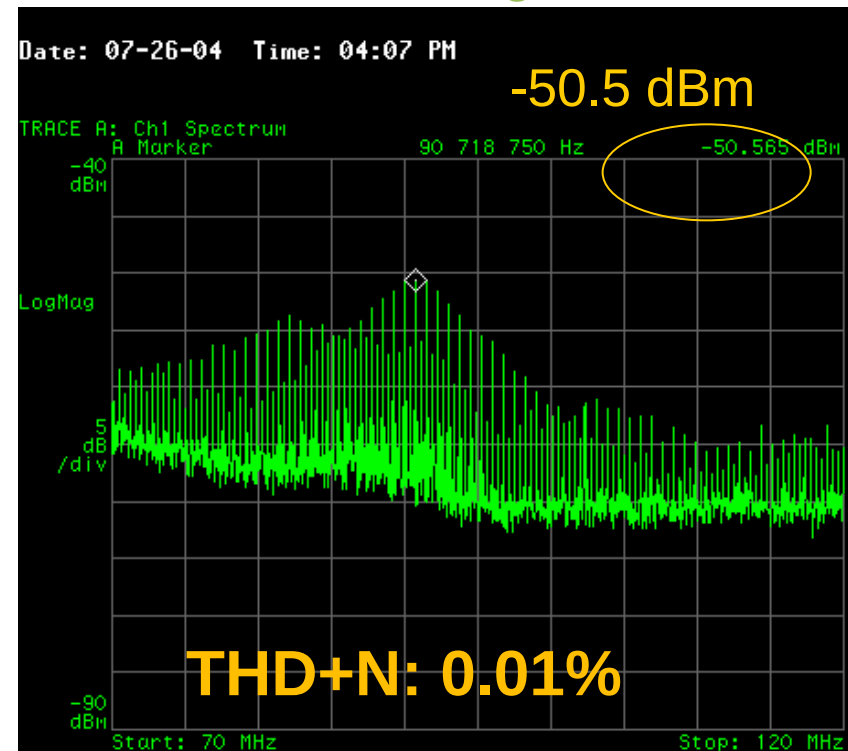
**High PSRR, substantially lower
intermodulation spectrum**

FSA95601 Feedback Benefits: Lower EMI in FM Band Will Improve with new Modulator and Power Stage Chips

Fast Switching FET



Slow Switching FET

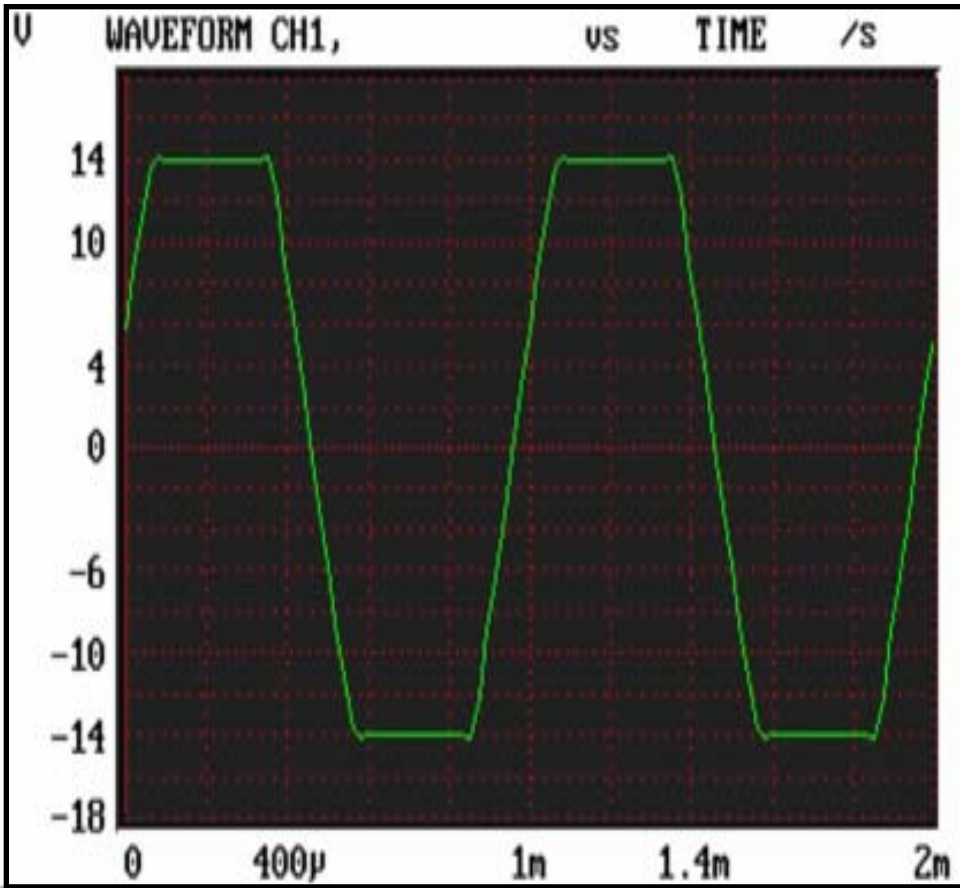


- EMI spectrum in FM band (70-120 MHz) has been reduced by 20 dB with slower switching power FETs
- Freescale Digital Amplifier Feedback Loop ensures minimal audio degradation
- EMI spectrum measurements done using "Artificial Network" recommended by CISPR25
- EMI source is mainly conductive

FSA95601 - Additional EMI Reduction Features

- **Programmable switching phasing of the 6 PWM channels**
- **PWM outputs have programmable driving strength**
- **Multiple controllers can synchronize their phasing to reduce EMI in systems with more than 6 channels**
- **Software control of main PWM switching frequency to overcome AM band interference**
- **Suppressed-carrier mode: Switching frequency plus odd harmonics are differentially cancelled across the load (speaker)**

Output Scope Plot with Saturation



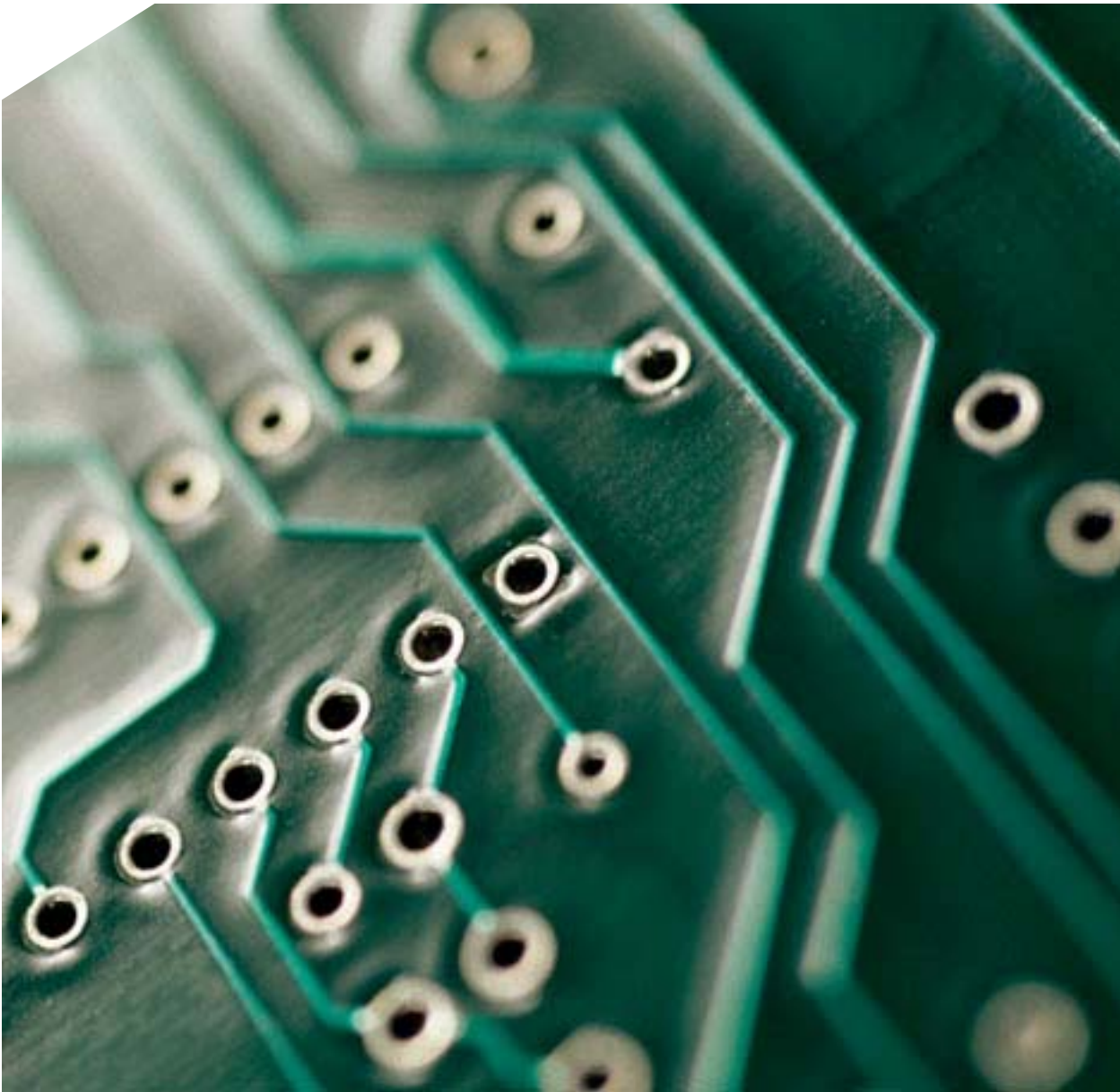
- PWM output capable of 100% modulation duty ratio maximizes power output capability
- Implements graceful transitions into and out of 0% duty ratio and 100% duty ratio

Variable Loop Order:

- Loop order is reduced when loop components saturate Integrators ADC

Variable Sample Point:

- Dynamic adjustment of ADC sample time

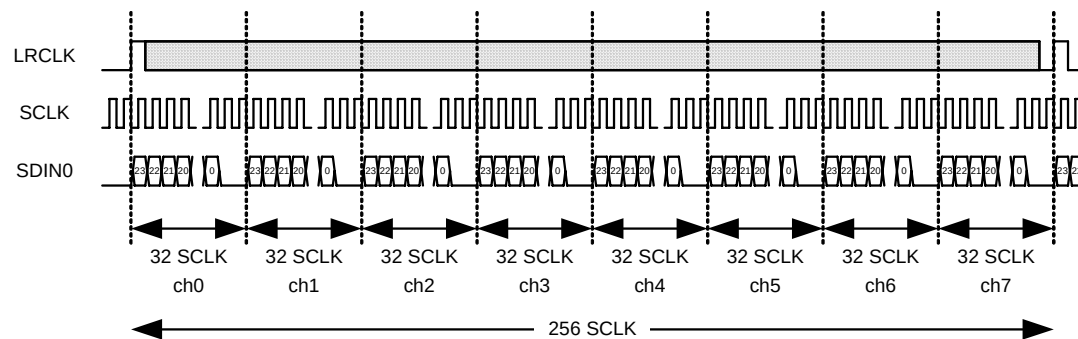
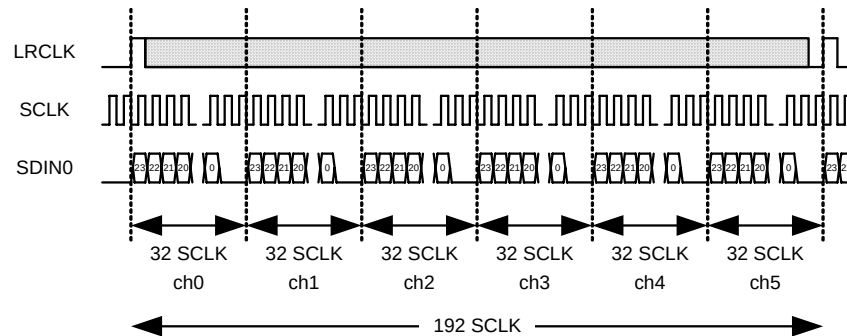


FSA95601 Peripherals

- **Serial digital audio interface for up to six input channels**
 - Supports 32 kHz to 192 kHz input audio sample rates
 - Supports four stereo serial input formats:
 - > 24-bit left justified
 - > 24-bit left justified delayed by 1 clock (I²S)
 - > 16-bit right justified
 - > 24-bit right justified
 - Supports two TDM serial input formats:
 - > 8 slots with 256Fs or 6 slots with 192Fs serial bit clock
 - Does not require master clock (ASRC built in)
 - Quiet Detection Mode (Mutes output when detecting continuous stream of input 0s)
 - De-emphasis IIR filter

TDM Serial Audio Interface

- Capable of both $192 F_s$ and $256 F_s$ modes



- Supports I²C or SPI mode
- Two I²C address pins allow up to four FSA95601 devices on the same I²C bus
- Auto increment mode provided for sending a series of data words efficiently

- Highly programmable volume control
- +18 dB to –96 dB range in 1/8 dB steps
- Independent volume control for each channel
- Master volume control
- Immediate or delayed adjustment modes
- Ramp mode with adjustable rate of volume change
- Zero Crossing mode adjusts at next zero
- Zero Crossing mode and ramp can be combined
- Variable soft/hard mute options
- Quiet enable option for digital silence

FSA95601 Asynchronous Sample Rate Converter

- **Programmable upsampler**
 - Upsamples audio data to 705.6 kHz or 768 kHz
 - Choice of 'steep' or 'gentle' upsampling filter
 - Automatic selection of optimal oversampling filter stages
 - Automatic muting when switching sampling frequencies
- **De-emphasis filter**
 - Can be used with any input sample rate (32kHz to 192 kHz)
- **Asynchronous Sample Rate Converter (ASRC)**
 - Allows synchronization to the fixed output rate
 - Can be bypassed if not needed

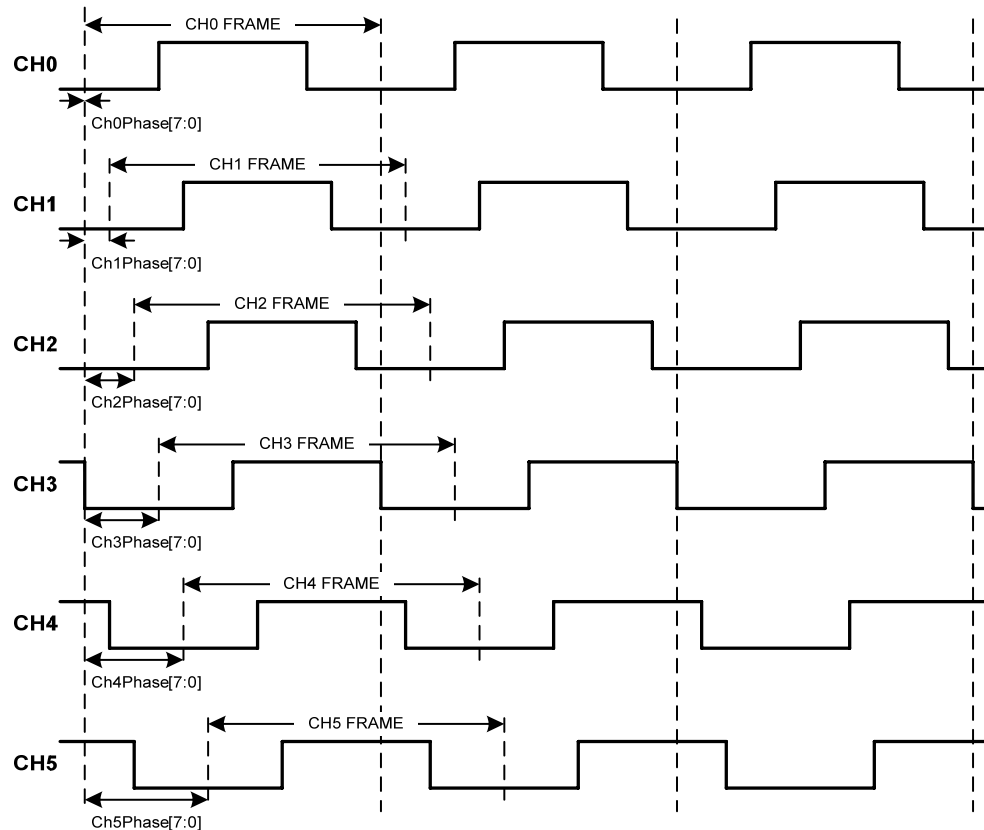
FSA95601 On-Chip Clock Generator

- **On-chip crystal oscillator**
 - Supports ~12 MHz, ~24 MHz, and ~48 MHz crystals
 - Output option to drive other devices
- **On-chip low-noise and low-jitter PLL**
 - Fully integrated
 - Provides high-quality PWM quantization clock for high fidelity PWM output
 - Option to use ~48 MHz crystal for quantization clock directly

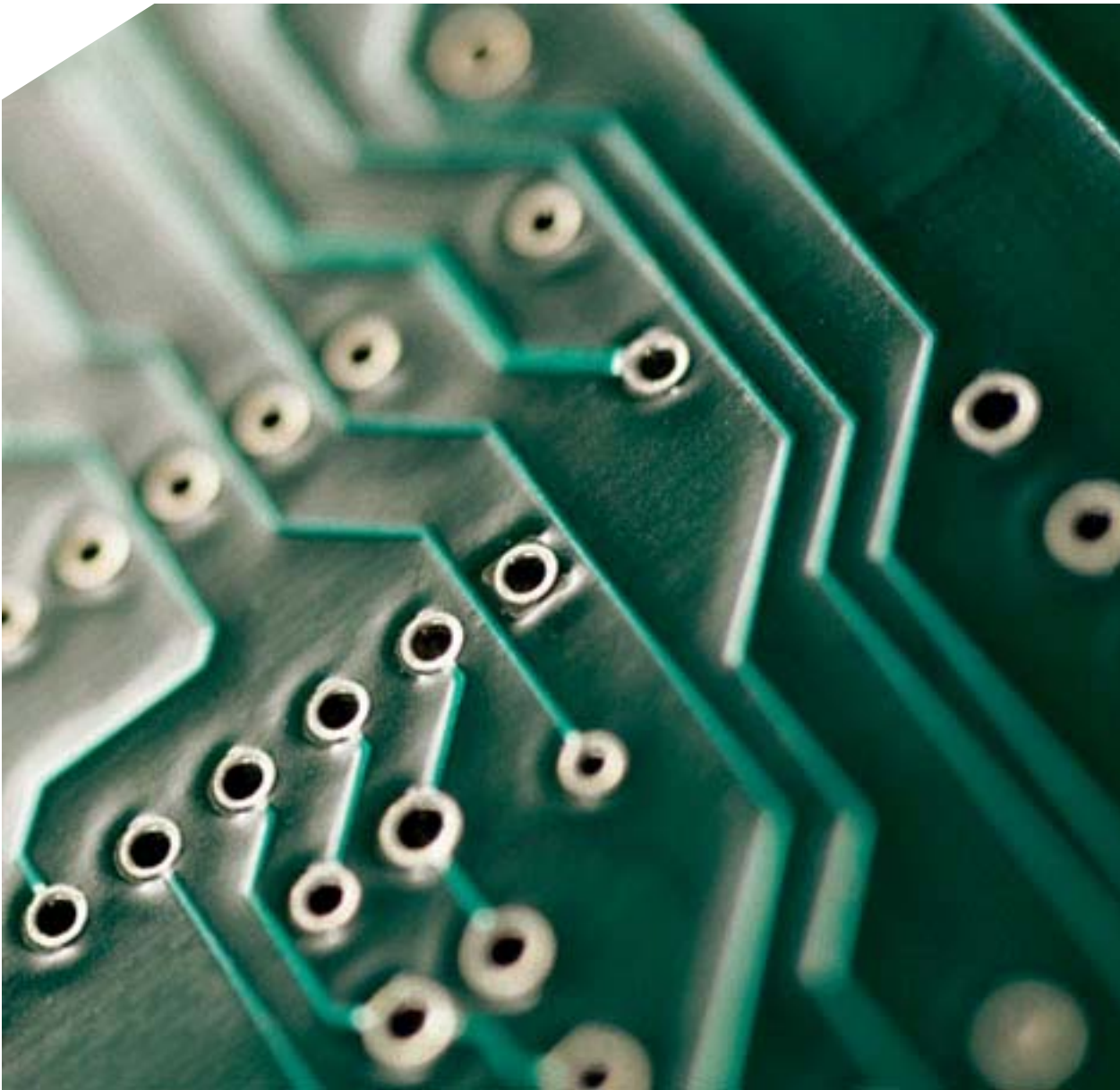
- **Programmable high fidelity PCM to PWM conversion**
 - Suppressed Carrier and Standard PWM options
- **High Performance Digital Feedback**
- **Feedback loop delay compensation to match feedback to backend layout and component delays**
- **Adjustable Break Before Make controls**
- **Programmable output driver strength to minimize EMI**

FSA95601 Programmable Phasing of PWM Output Channels

Programmable phase control in 20 ns increments



- **Switching frequency plus odd harmonics are differentially cancelled across the load (speaker)**
- **Allows smaller LC filter**
 - Reduced order filter
 - Smaller inductance
- **Reduced losses at low volume**



Development Tools

DAEVM601 Digital Amplifier EVM

Speaker Terminals
2-8 Ohm typical

Low Pass
Filter Caps (C)
4 per Ch.

Dual Core
Inductor (L)
1 per Ch.

Discrete FET
H-Bridge
4 FETs per Ch.

FET Line
Driver

Power stage
supply voltage
10V-18V

PWM and Feedback
Signal Header
Can connect to alternative
power stage daughter cards

FSA95601

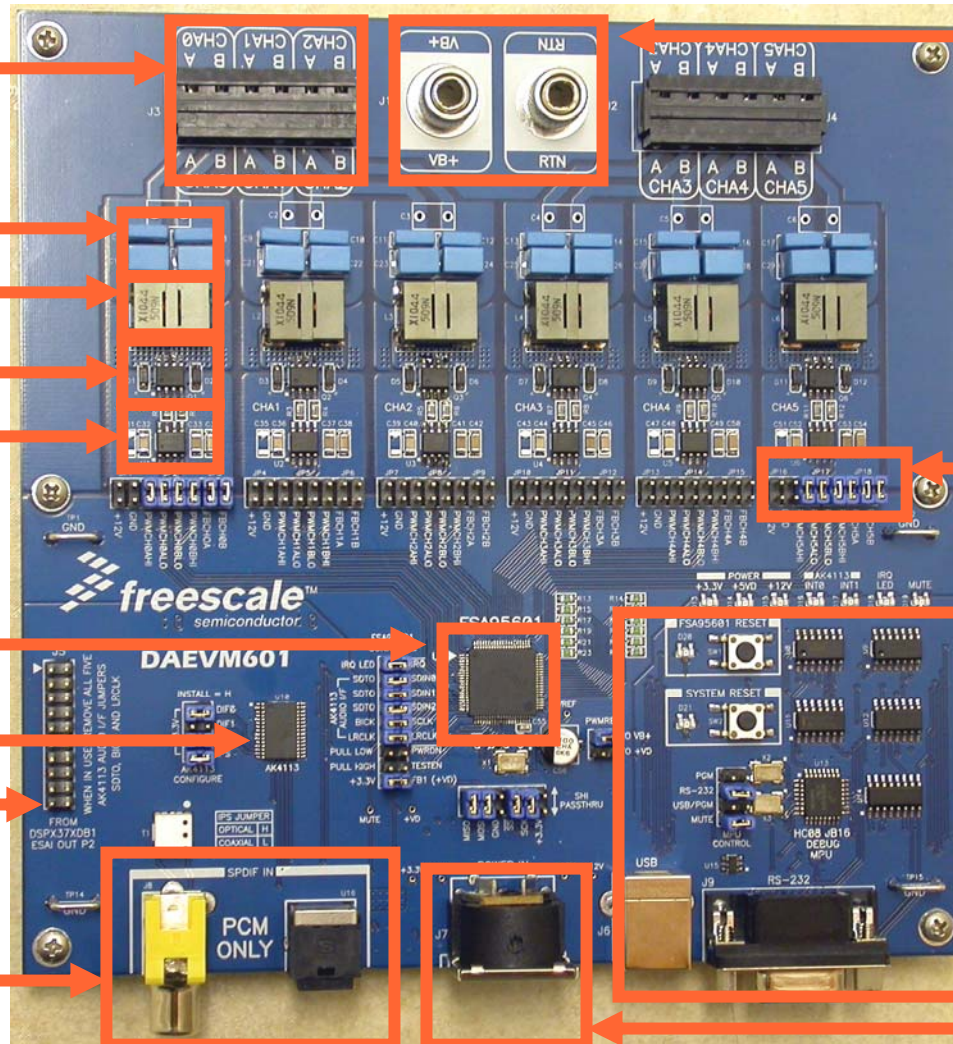
SPDIF Receiver

Multi-Channel
Audio interface
Header to connect
To FSL DSP EVM

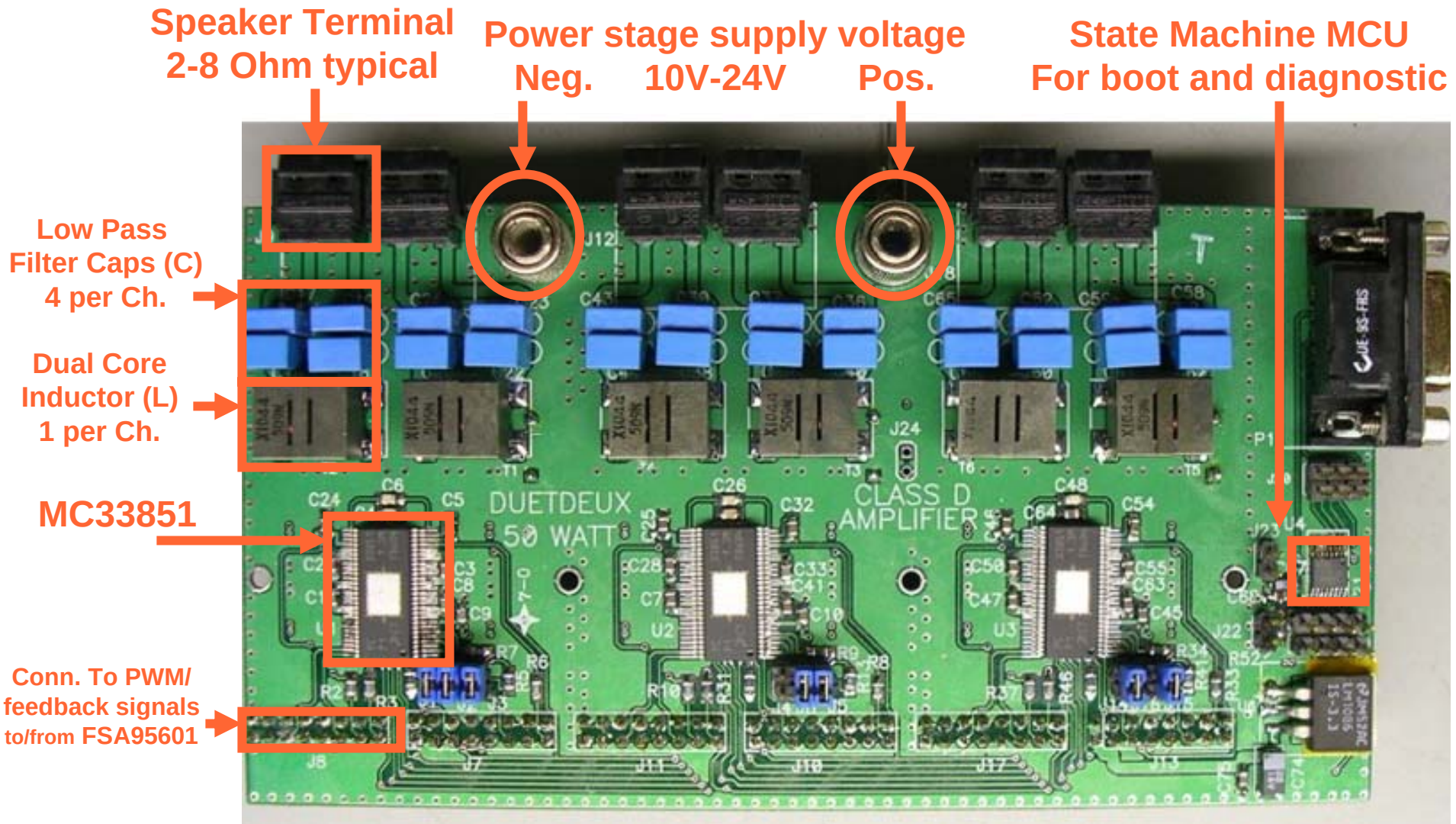
Optical/Electrical
SPDIF input

SDI Debugger Interface
Emulates SPI/I2C
communication

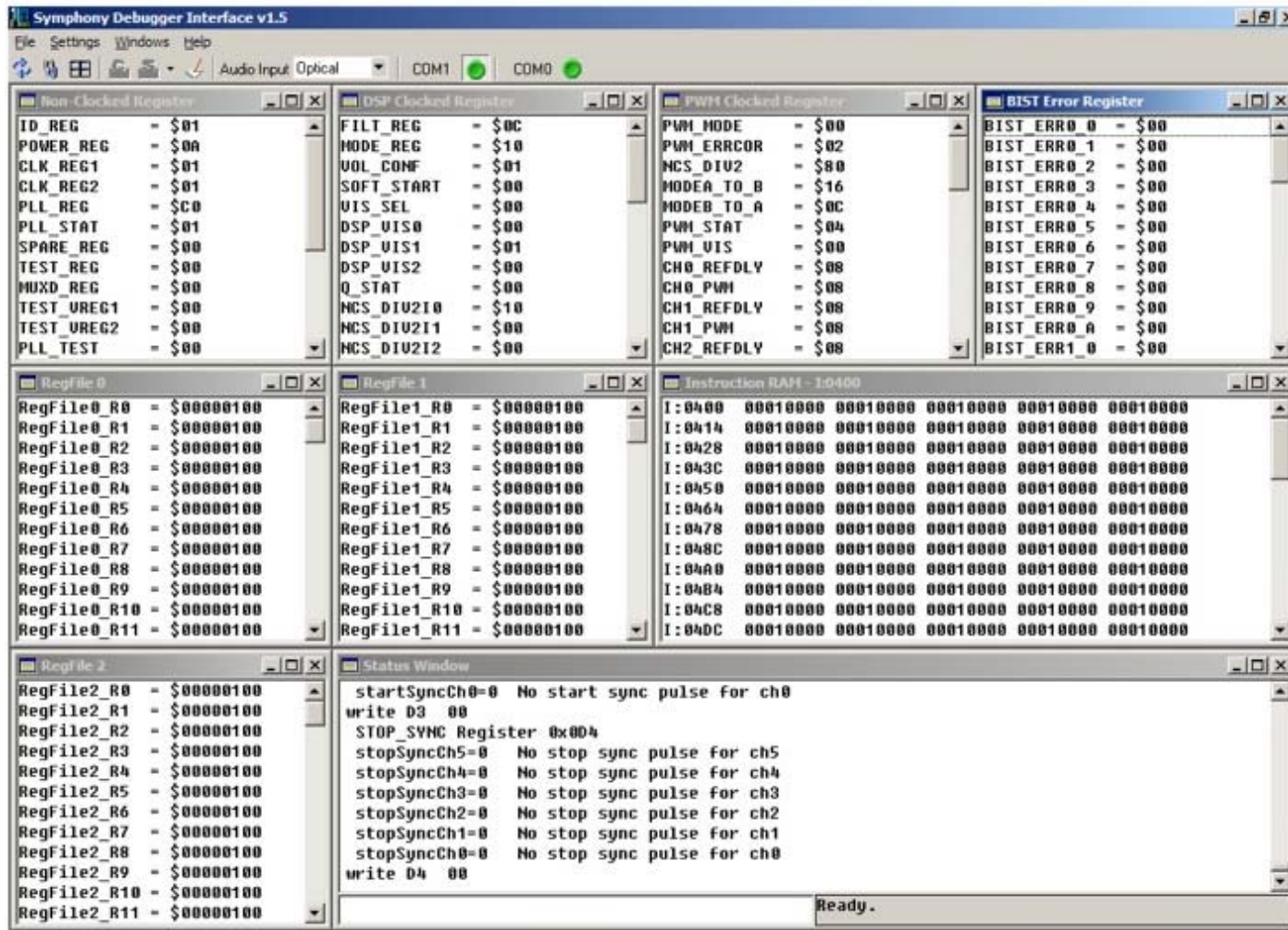
Digital Power
Supply Connector



MC33851 Daughter card



SDI Debugger

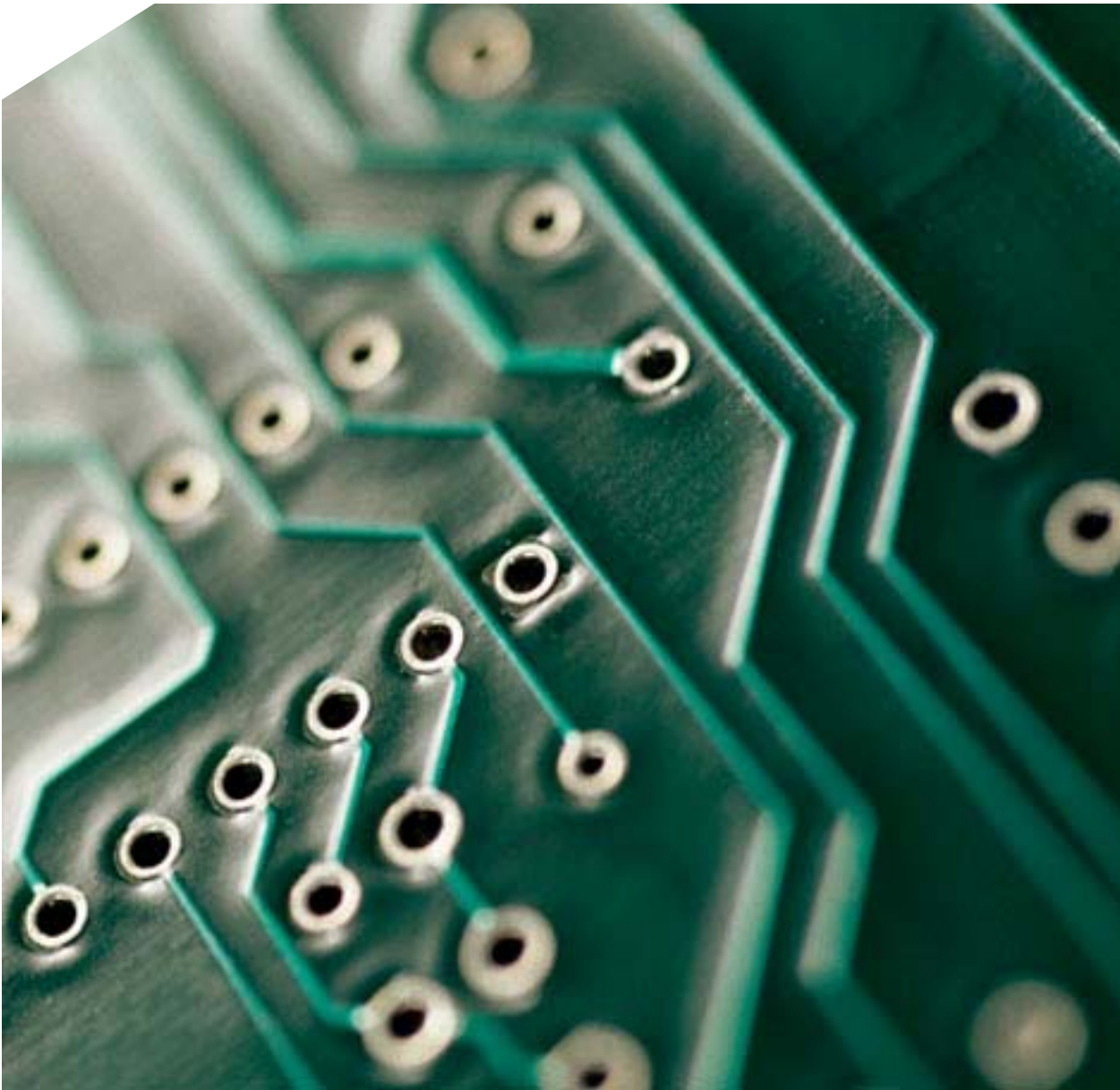


SDI Debugger

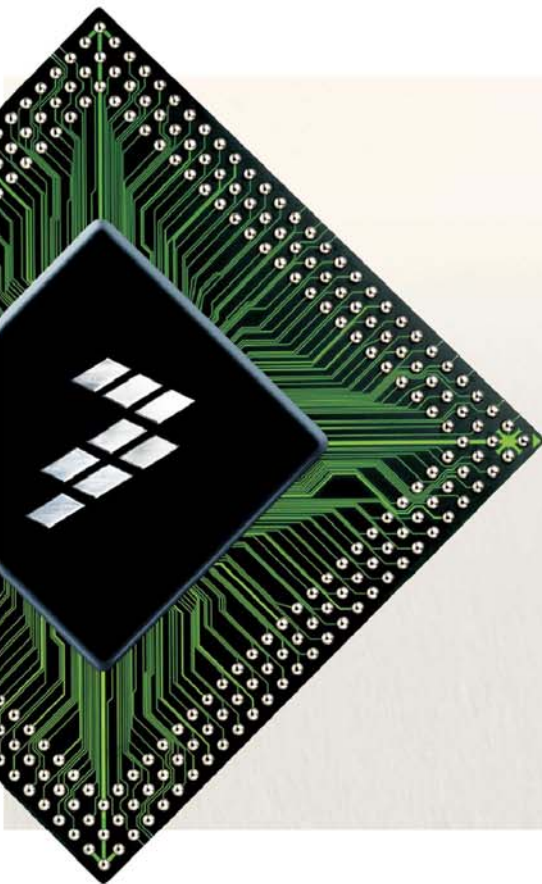
Allows emulation of control MCU for SPI/I²C communication to FSA95601.

Sample macros available to boot FSA95601 in typical modes. Command line allows direct real time control of variables such as volume.

- **Digital (switching)** Class D Technology provides increased power output with lower heat dissipation allowing **smaller form factors**
- Superior power supply noise rejection via **High Performance Digital Feedback** that substantially reduces power supply costs
- Outstanding **SNR and THD**
- Practical **solutions to overcome potential EMI issues** associated with switching amplifier technology
- Advanced **diagnostics & protection**
- Increased power output with **lower heat dissipation** allowing more channels in smaller form factors
- Sound fidelity **comparable or better** to existing analog designs
- Single semiconductor supplier offering complete **system solution**, from DSP processing to switching amplifier modulator to power stages
- **Established market presence** in Automotive, Consumer and Professional Multi-Channel Digital Audio Processing



Q & A



Freescale Technology Forum

Design Freedom.